

N. GREGORY

MANKIWI

PRINCIPLES OF

ECONOMICS

Eighth Edition



CHAPTER

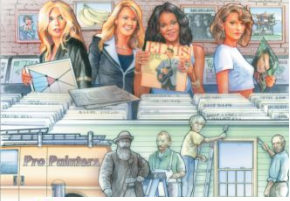
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Application: International Trade

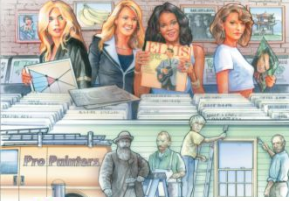
Premium PowerPoint Slides by:
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Look for the answers to these questions:

- What determines how much of a good a country will import or export?
- Who benefits from trade? Who does trade harm? Do the gains outweigh the losses?
- If policymakers restrict imports, who benefits? Who is harmed? Do the gains from restricting imports outweigh the losses?
- What are some common arguments for restricting trade? Do they have merit?



- A country has a comparative advantage in a good
 - If it produces the good at lower opportunity cost than other countries
- Countries can gain from trade
 - If each exports the goods in which it has a comparative advantage



The Determinants of Trade

- The equilibrium without trade
 - Only domestic buyers and sellers
 - Equilibrium price and quantity
 - Determined on the domestic market
 - Total benefits
 - Consumer surplus
 - Producer surplus



World Price and Comparative Advantage

- P_W = the world price of a good, the price that prevails in world markets
- P_D = domestic price without trade
- If $P_D < P_W$,
 - Domestic country has comparative advantage, country exports the good
- If $P_D > P_W$,
 - Domestic country does not have comparative advantage, country imports the good



The Small Economy Assumption

- A small economy is a price taker in world markets:
 - Its actions have no effect on P_W
 - When a small economy engages in free trade, P_W is the only relevant price:
 - No seller would accept less than P_W (can sell the good for P_W in world markets)
 - No buyer would pay more than P_W (can buy the good for P_W in world markets)

A Country That Exports Soybeans

Without trade,

$$P_D = \$4$$

$$Q = 500$$

$$P_W = \$6$$

Under free trade,

domestic

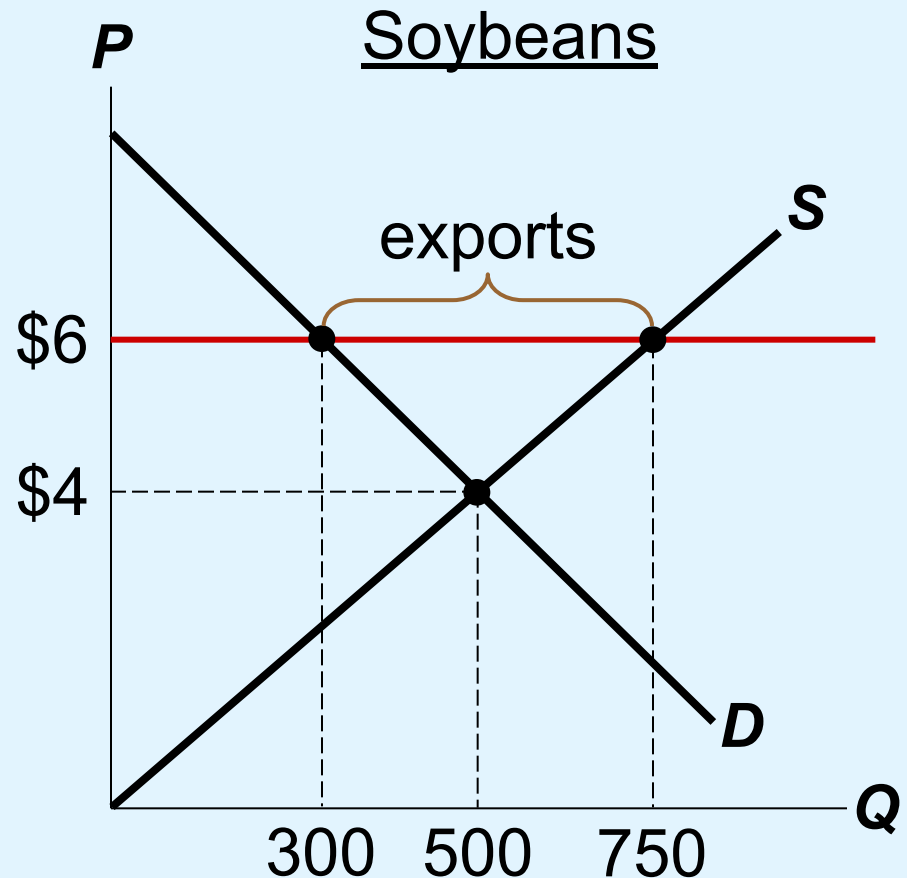
consumers

demand 300

domestic producers

supply 750

exports = 450



A Country That Exports Soybeans

Without trade,

$$CS = A + B$$

$$PS = C$$

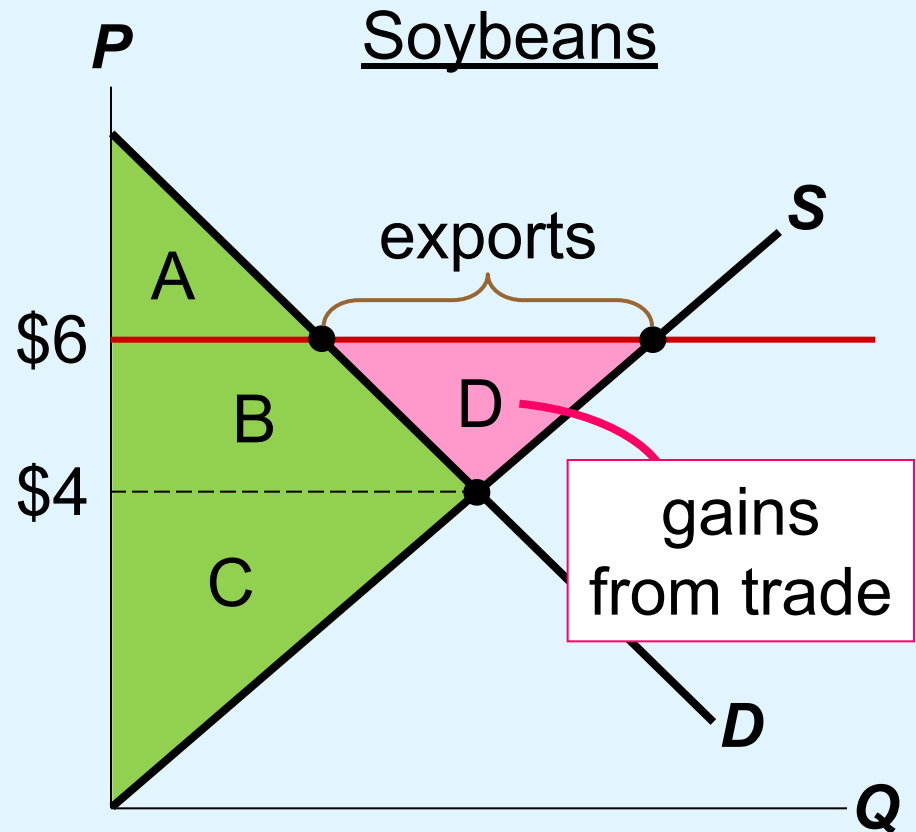
$$\begin{aligned}\text{Total surplus} \\ &= A + B + C\end{aligned}$$

With trade,

$$CS = A$$

$$PS = B + C + D$$

$$\begin{aligned}\text{Total surplus} \\ &= A + B + C + D\end{aligned}$$



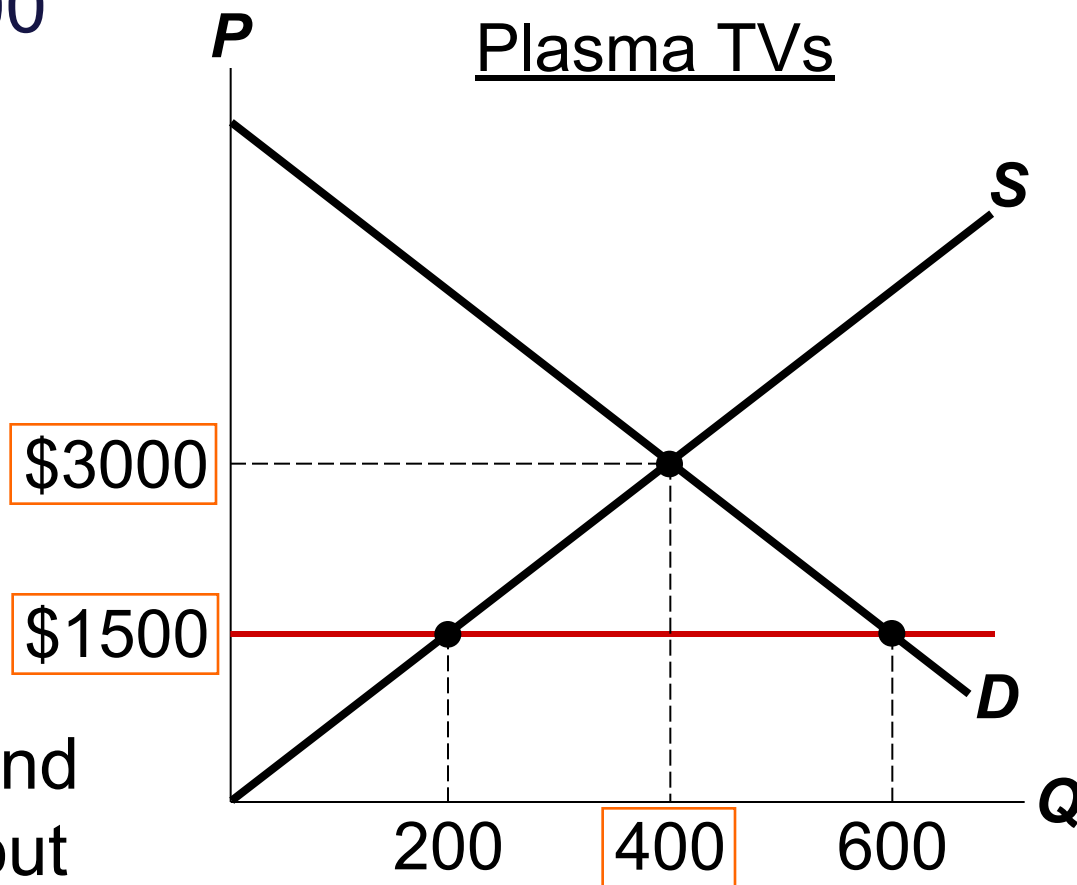
Active Learning 1

Analysis of Trade

Without trade,
 $P_D = \$3000$, $Q = 400$

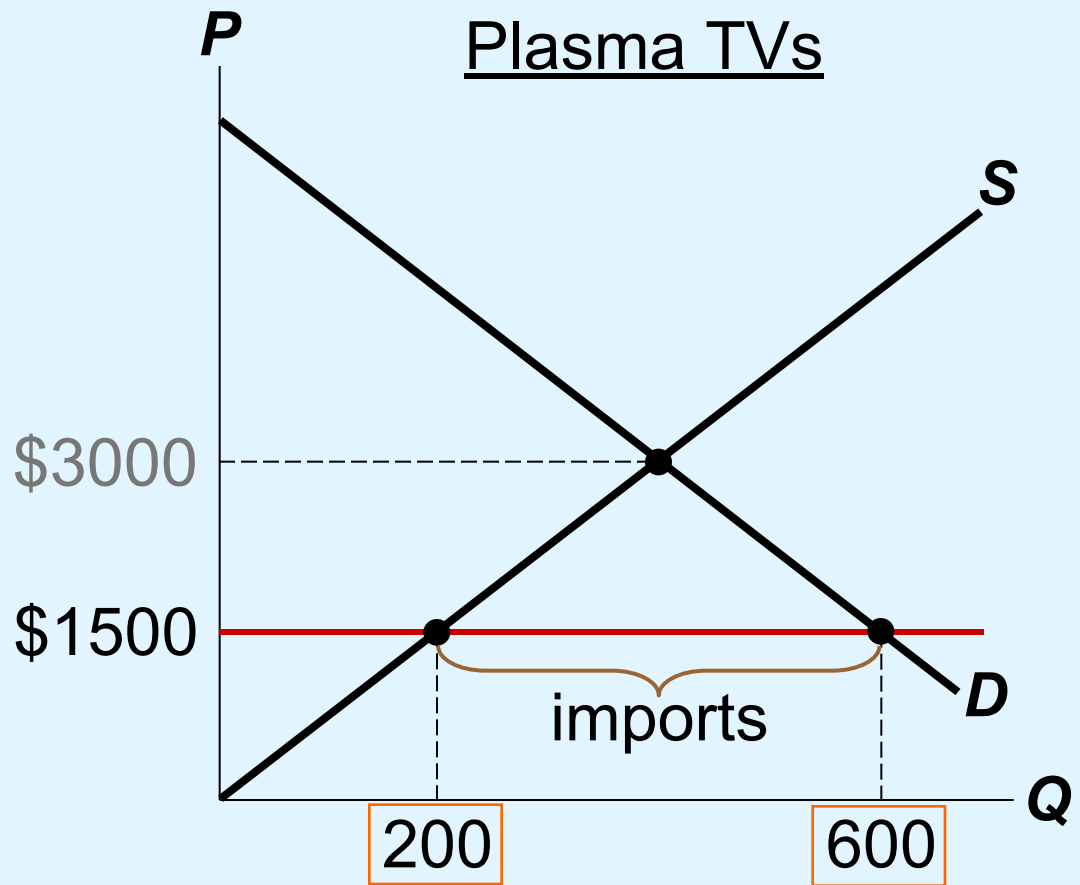
In world markets,
 $P_W = \$1500$

- Under free trade, how many TVs will the country import or export?
- Identify CS, PS, and total surplus without trade, and with trade.



Under free trade,

- domestic consumers demand 600
- domestic producers supply 200
- imports = 400



Without trade,

$$CS = A$$

$$PS = B + C$$

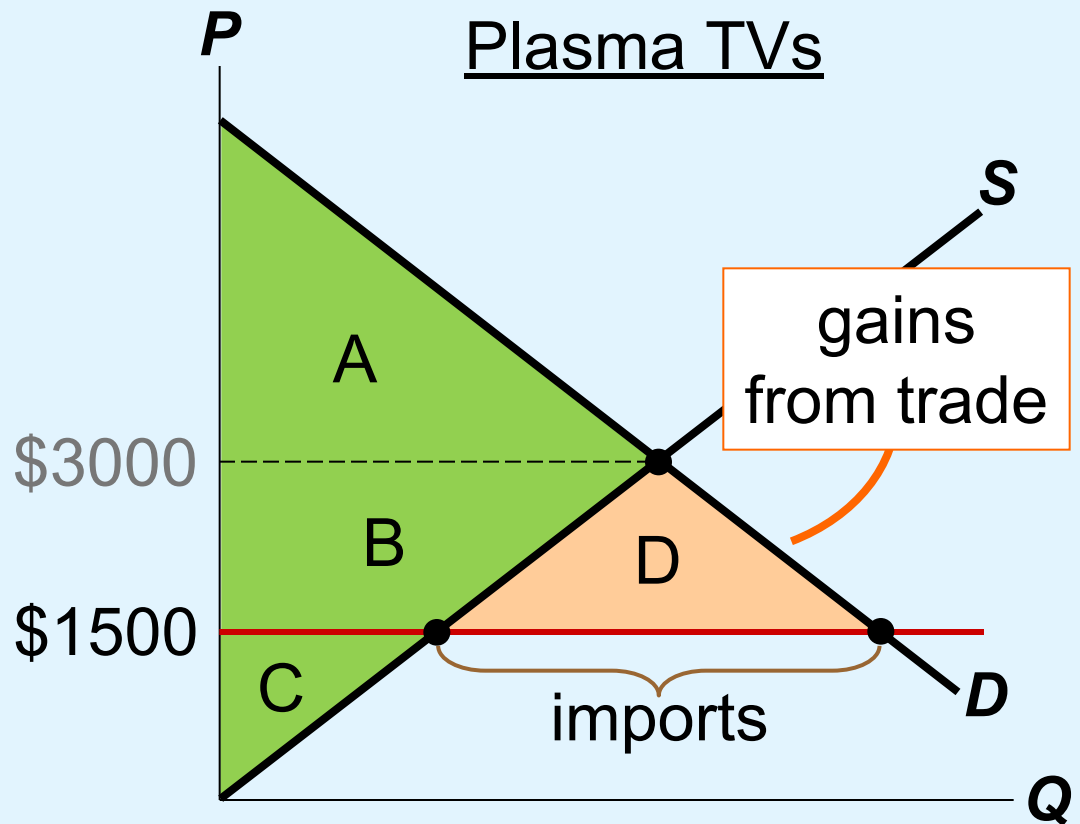
$$\begin{aligned} \text{Total surplus} \\ = A + B + C \end{aligned}$$

With trade,

$$CS = A + B + D$$

$$PS = C$$

$$\begin{aligned} \text{Total surplus} \\ = A + B + C + D \end{aligned}$$



Summary: The Welfare Effects of Trade

	$P_D < P_W$	$P_D > P_W$
direction of trade	exports	imports
consumer surplus	falls	rises
producer surplus	rises	falls
total surplus	rises	rises

*Whether a good is imported or exported,
trade creates winners and losers.
But the gains exceed the losses.*



Winners and Losers From Trade

- Other benefits of international trade
 - Consumers: increased variety of goods
 - Producers: lower costs - economies of scale
 - Increased competition: reduce market power of domestic firms (increase total welfare)
 - Enhanced flow of ideas, facilitates the spread of technological advances around the world



Winners and Losers From Trade

- Then why all the opposition to trade?
 - The losers have more incentive to organize and lobby for restrictions on trade:
 - Losses: concentrated among a small group of people, who feel them acutely
 - Gains: spread thinly over many people, who may not see how trade benefits them
 - The winners from trade could compensate the losers and still be better off (such compensation rarely occurs)



Winners and Losers From Trade

- Tariff
 - Tax on goods produced abroad and sold domestically
- Free trade
 - Domestic price = World price
- Tariff on imports
 - Raises domestic price above world price
 - By the amount of the tariff

Analysis of a Tariff on Cotton Shirts

$$P_w = \$20$$

Free trade:

buyers demand 80

sellers supply 25

imports = 55

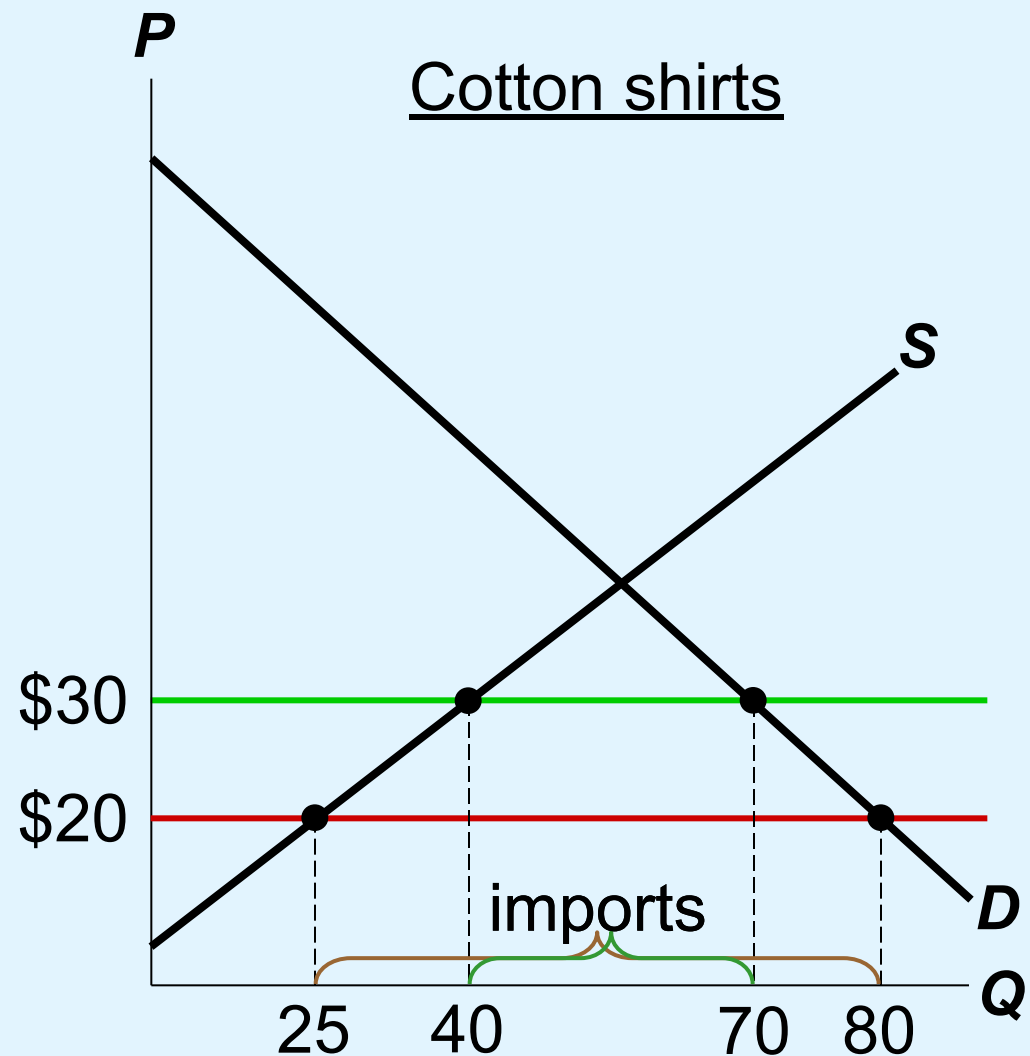
$$\underline{T = \$10/\text{shirt}}$$

price rises to \$30

buyers demand 70

sellers supply 40

imports = 30



Analysis of a Tariff on Cotton Shirts

Free trade

$$CS = A + B + C + D + E + F$$

$$PS = G$$

$$\text{Total surplus} = A + B + C + D + E + F + G$$

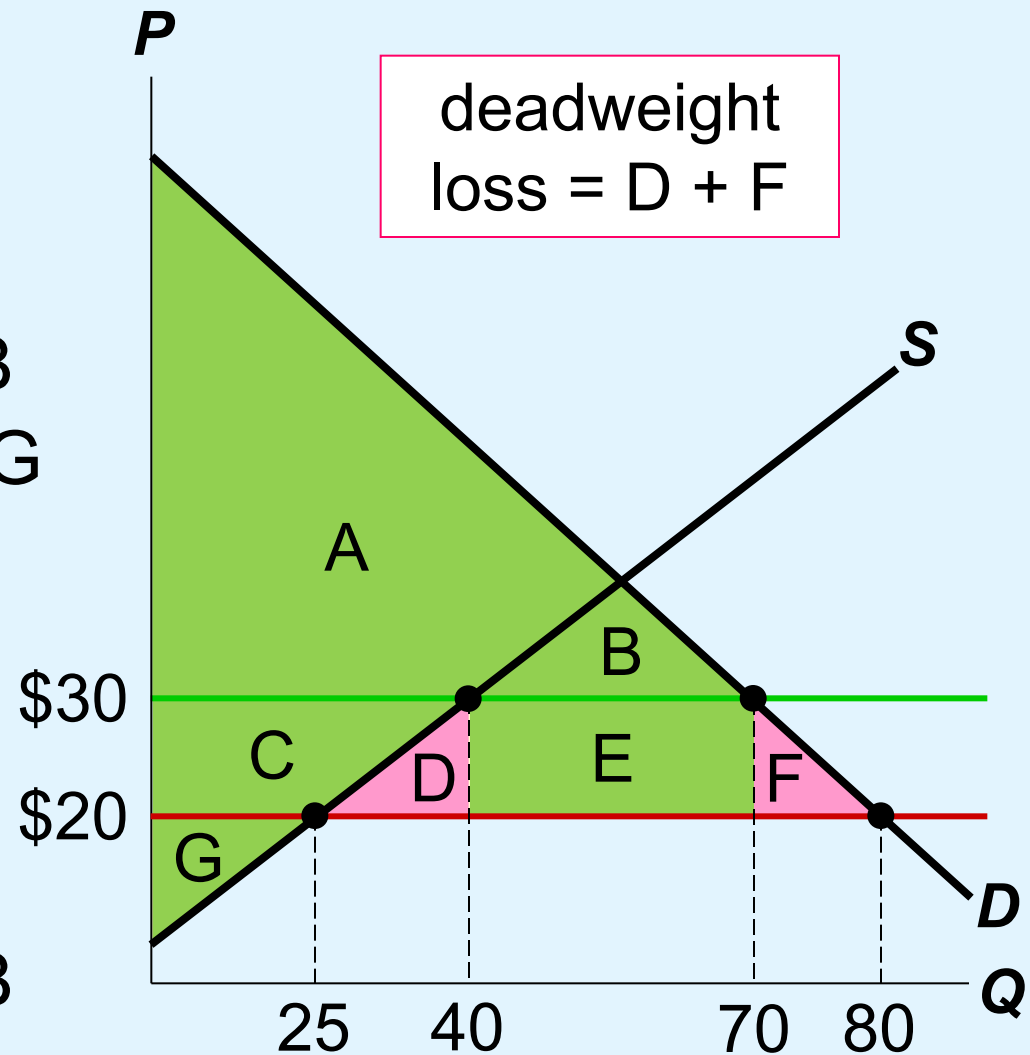
With tariff

$$CS = A + B$$

$$PS = C + G$$

$$\text{Revenue} = E$$

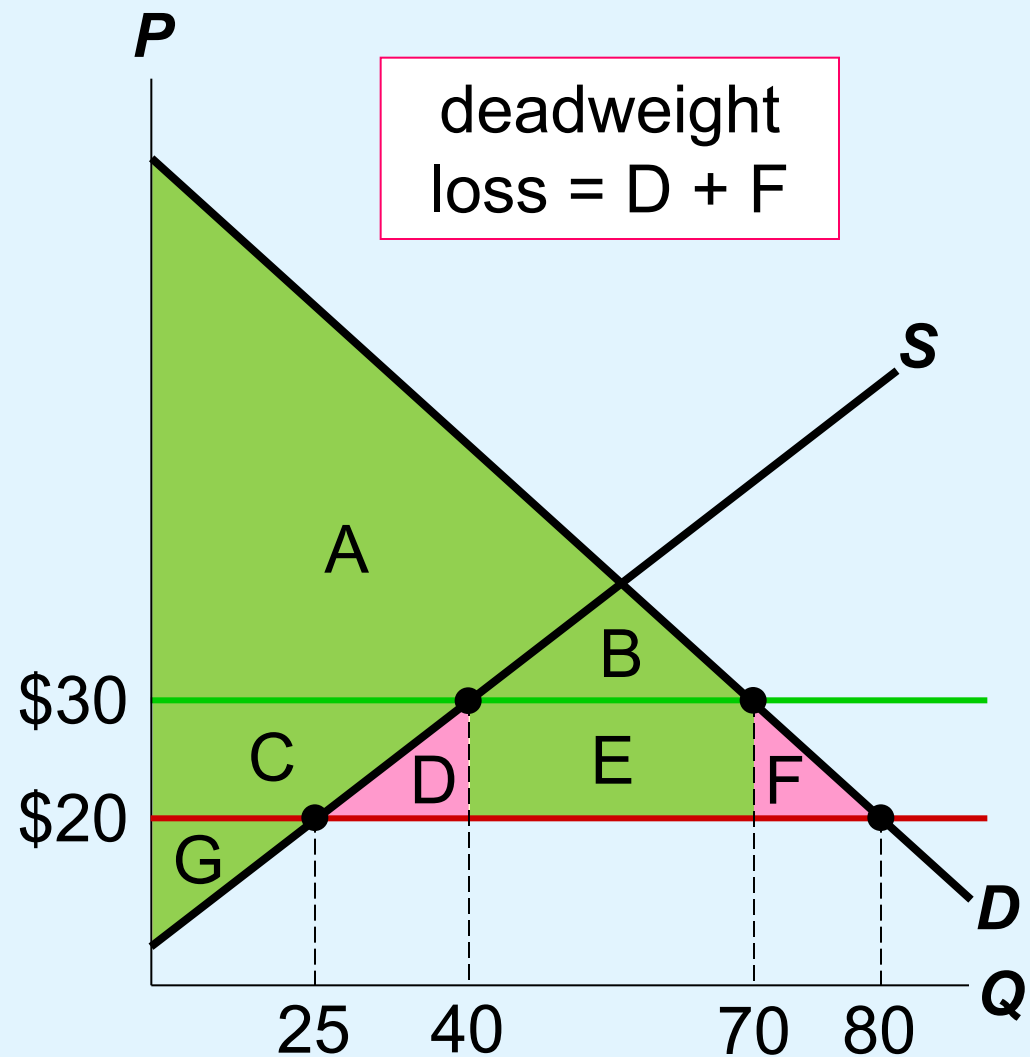
$$\text{Total surplus} = A + B + C + E + G$$

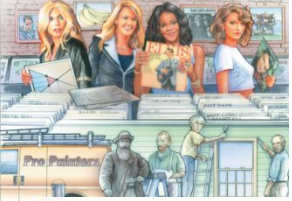


Analysis of a Tariff on Cotton Shirts

D = deadweight loss from the overproduction of shirts

F = deadweight loss from the under-consumption of shirts

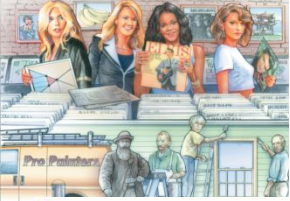




Import Quotas

- Import quota

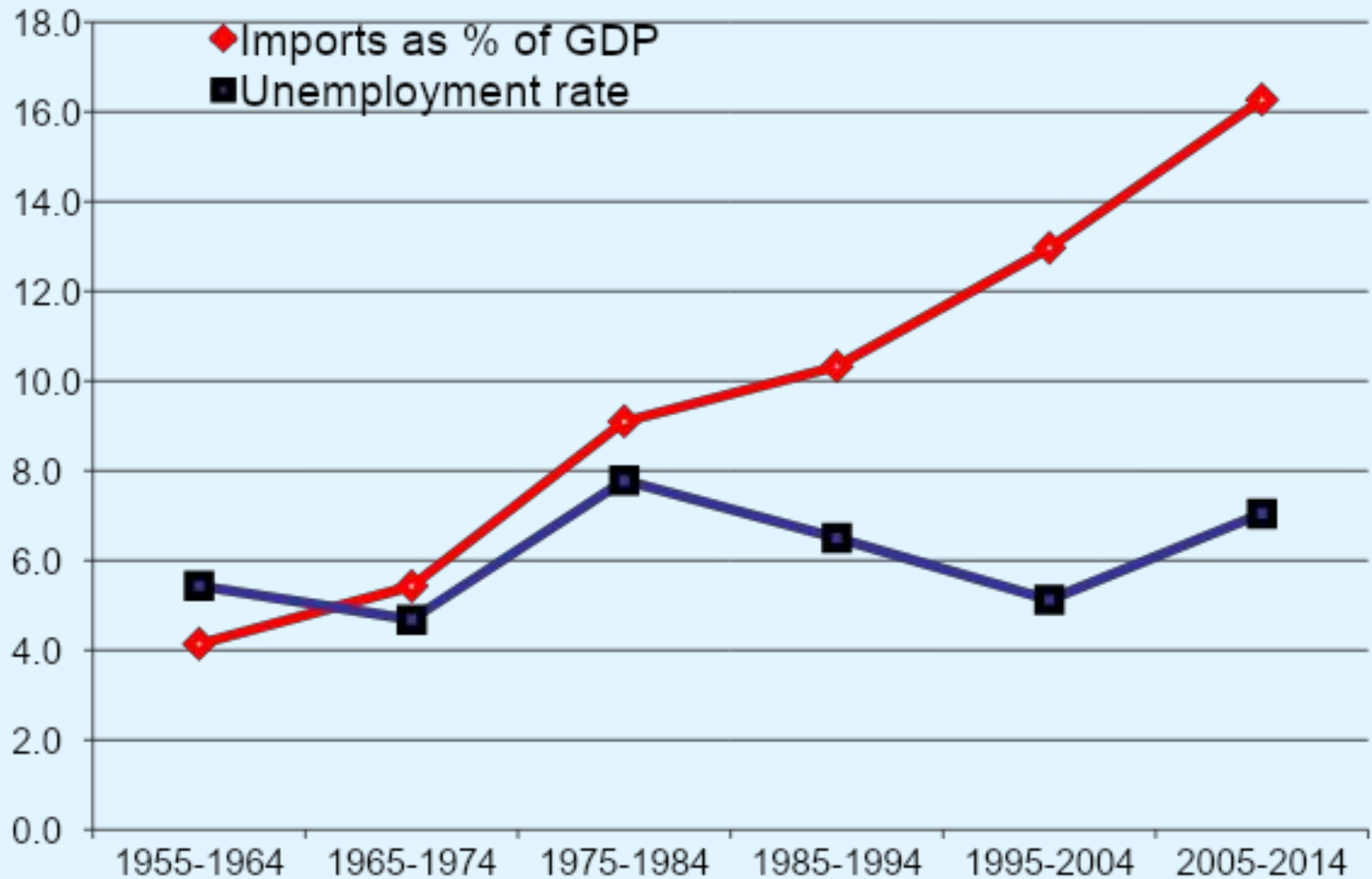
- Quantitative limit on imports of a good
- Mostly has the same effects as a tariff:
 - Raises price, reduces quantity of imports
 - Reduces buyers' welfare
 - Increases sellers' welfare
- Creates profits for the foreign producers of the imported goods, who can sell them at higher price

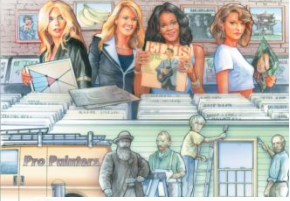


Arguments For Restricting Trade

- The jobs argument
 - “Trade with other countries destroys domestic jobs”
 - Free trade creates jobs at the same time that it destroys them
 - Total unemployment does not rise as imports rise, because job losses from imports are offset by job gains in export industries....

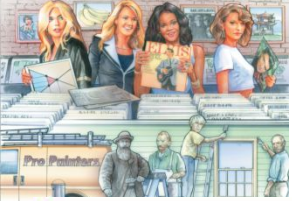
U.S. Imports & Unemployment, decade averages, 1955–2014





Arguments For Restricting Trade

- The national-security argument
 - “The industry is vital for national security and it should be protected from foreign competition, to prevent dependence on imports that could be disrupted during wartime”
 - When there are legitimate concerns over national security
 - But producers may exaggerate their own importance to national security to obtain protection from foreign competition



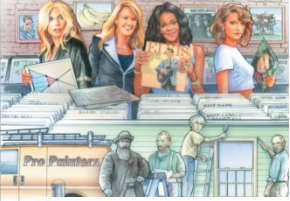
Arguments For Restricting Trade

- The infant-industry argument
 - “New industries need temporary trade restriction to help them get started”
 - Difficult to implement in practice
 - The temporary policy is hard to remove
 - Protection is not necessary for an infant industry to grow



Arguments For Restricting Trade

- The unfair-competition argument
 - “Producers argue their competitors in another country have an unfair advantage, e.g. due to government subsidies”
 - Increase in total surplus for the country
 - We should welcome imports of low-cost products subsidized by the other country’s taxpayers
 - The gains to our consumers will exceed the losses to our producers



Arguments For Restricting Trade

- The protection-as-a-bargaining-chip argument
 - “Trade restrictions can be useful when we bargain with our trading partners”
 - The threat may not work



- World Trade Organization, WTO
- Unilateral approach to achieve free trade
 - Remove its trade restrictions on its own
 - Great Britain, 19th century
 - Chile and South Korea, recent years
- Multilateral approach to free trade
 - Reduce its trade restrictions while other countries do the same
 - NAFTA, GATT

- North American Free Trade Agreement (NAFTA)
 - 1993, lowered trade barriers among the United States, Mexico, and Canada
- General Agreement on Tariffs and Trade (GATT)
 - Continuing series of negotiations among many of the world's countries with the goal of promoting free trade



- **GATT**
 - United States helped to found GATT
 - After World War II
 - In response to the high tariffs imposed during the Great Depression
 - Successfully reduced the average tariff among member countries from about 40% to 5%
 - Enforced by the WTO
 - 2015: 162 countries; more than 97 % of world trade



- **Advantages of the multilateral approach**
 - Potential to result in freer trade than unilateral approach
 - Reduce trade restrictions abroad and at home
 - Political advantage
 - Producers are fewer and better organized than consumers
 - Greater political influence

Summary

- A country will export a good if the world price of the good is higher than the domestic price without trade. Trade raises producer surplus, reduces consumer surplus, and raises total surplus.
- A country will import a good if the world price is lower than the domestic price without trade. Trade lowers producer surplus but raises consumer and total surplus.
- A tariff benefits producers and generates revenue for the government, but the losses to consumers exceed these gains.

Summary

- Common arguments for restricting trade include: protecting jobs, defending national security, helping infant industries, preventing unfair competition, and responding to foreign trade restrictions.
- Some of these arguments have merit in some cases, but economists believe free trade is usually the better policy.